AGNES KUOK, SARA HONG, SARA WIN, SOPHIA CHIO, YOHANNA LEI

SACRED HEART CANOSSIAN COLLEGE (ENGLISH SECTION)

Macau Biodiversity Data Science Challenge

ASSESSING THE IMPACT OF ENVIRONMENTAL AND ANTHROPOGENIC FACTORS ON MACAU'S BIODIVERSITY - A FOCUS ON MOUNTAINOUS ECOSYSTEMS AND STRATEGIES FOR SUSTAINABLE CONSERVATION -

ABSTRACT

Urban ecosystems face **mounting biodiversity threats**, yet **mountainous regions** within cities—such as Macau's hills—remain **understudied**. This project investigates biodiversity across three hills in Macau: **Grand Taipa National Hill, Colina da Guia (Guia Hill) and Alto de Coloane**, by designing a **Predictive EcoHealth and Biodiversity model (PEB model)** integrating **climatic**, **geographic**, and **anthropogenic factors**. Using **field surveys** and **engineered Arduino sensors**, we collected data on **soil quality** (e.g., pH, organic content), **water systems** (e.g., dissolved oxygen, turbidity) and **human impacts** (e.g., noise level, road density, invasive species). **Climate variables** (precipitation acidity, air quality indices) were **supplemented with governmental datasets**.

Our model **weights factors hierarchically**: climate (30%, e.g., temperature, storm frequency), geography (30%, e.g., altitude, soil quality), and human activity (40%, e.g., population density, invasive species). **Preliminary results reveal disparities** in biodiversity between hills, with one site showing **significantly higher species richness** correlated with low population density. **Dominant, rare and indicator species** were identified to assess ecosystem health.

The study highlights **specific conservation priorities**, such as **reducing nutrient runoff** and **mitigating noise pollution**, to enhance biodiversity. By **comparing** modeled outcomes with field observations, we **demonstrate the index's utility for urban biodiversity** management, emphasizing the **need for policy-driven habitat protection**.

INTRODUCTION

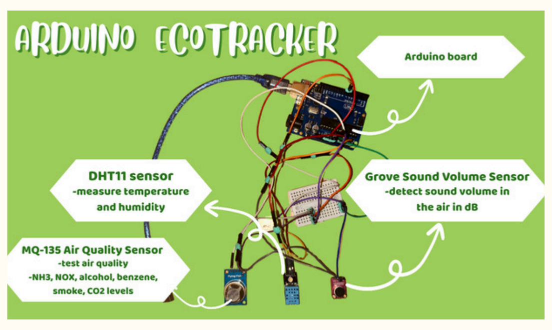
Biodiversity refers to the **variety of life** in a particular **habitat** or **ecosystem**, encompassing **species richness**, **genetic diversity**, and the **ecological interactions** that sustain biological communities. In Macao's predominantly urban landscape, **Coloane Hill**, **Taipa Hill** and **Guia Hill** stand out as the territory's **most ecologically intact terrestrial environments**. These elevated areas preserve **characteristic communities** of native flora, resident and migratory birds, and small mammals that collectively represent the **region's natural heritage**. Their **distinct topography** and **vegetation** structure create unique microhabitats that support these biological communities. As three of Macao's natural ecosystems, they serve as **essential reference sites** for developing ecological assessment models.

SCIENTIFIC OBJECTIVES

- Use **different indexes** to quantify species evenness and dominance for the Grand Taipa National Hill, Colina da Guia and Alto de Coloane.
- Develop **predictive ecological models** for extrapolation to other areas.
- **Identify key species** groups including dominant, rare, and indicator species of the three hills, to evaluate ecosystem health.
- **Visualize spatial distributions** of species to pinpoint conservation hotspots for rare taxa and invasion fronts.
- **Benchmark Macau's biodiversity** against other places, such as Hong Kong.

MATERIALS AND METHODS

1



Arduino Coding

2

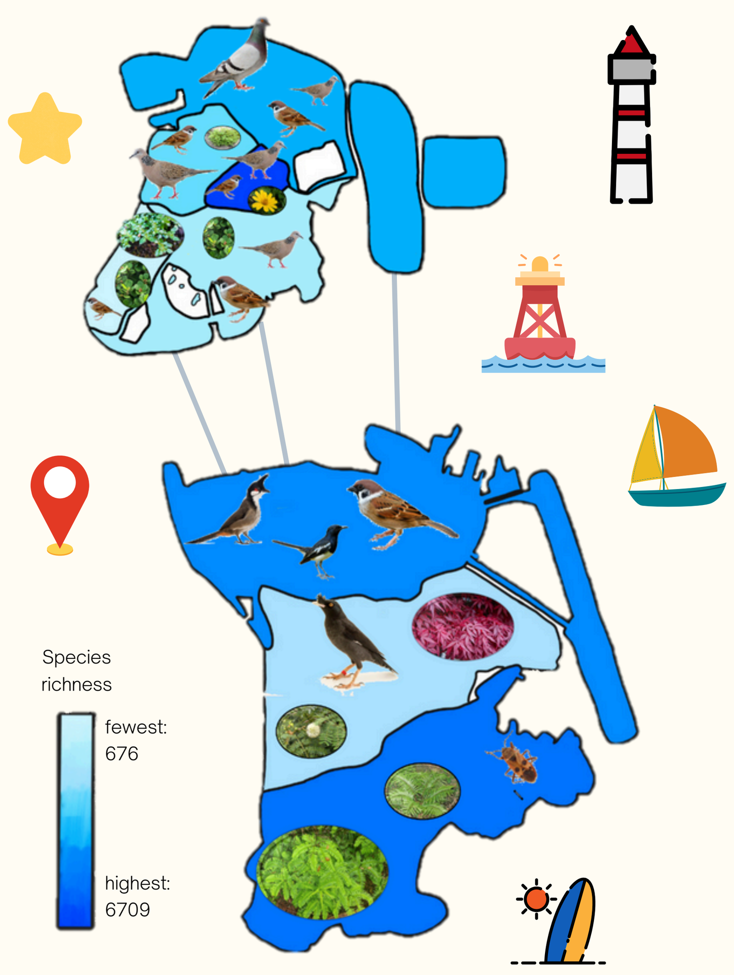
Calibration of pH meter using pH buffer solution → Measure the pH of the soil samples

3

Mix and stir the soil on a hot plate with potassium chloride solution → Filter the solution to extract a clear solution with the organic matter → Test the nitrate content using reagent test strips

SPATIAL DISTRIBUTION MAP OF SPECIES IN MACAO

Dominant species are shown in particular regions.



MODELLING AND ANALYSIS

The **Predictive EcoHealth and Biodiversity (PEB)** model quantifies **ecosystem health** through composite scores of **biodiversity potential**, **environmental suitability**, and **ecological integrity**. It integrates **weighted climatic** and **anthropogenic factors**, standardized via **min-max normalization**, to pinpoint conservation priorities.

Step 1 Identify Key Factors → Step 2 Factor Weighing → Step 3 Variable Normalisation (min-max scaling) → Step 4 Model Testing and Integration

Step 1 + 2

Table 1. Three main factors in PEB model

Score	Weighting
Climate (C)	30%
Geography (G)	30%
Human Activities (H)	40%

Table 2. Branching Factors of Climate

Factors		Weighting (α_i)
Precipitation (C_P)	Precipitation Acidity (C_{PA})	15%, $i = 1$
	Precipitation Volume (C_{PV})	15%, $i = 2$
Air Quality Metrics (C_{AQM})	Air Quality Index (C_{AQI})	15%, $i = 3$
	Humidity (C_H)	15%, $i = 4$
Mean Temperature (C_{MT})		15%, $i = 5$
Storm (C_S)	Storm Intensity (C_{SI})	10%, $i = 6$
	Storm Frequency (C_{SF})	15%, $i = 7$

Table 3. Branching Factors of Geography

Factors		Weighting (β_i)
Water System Metrics (G_{WSM})	Concentration of Dissolved Oxygen (G_{CDO})	15%, $i = 1$
	Nutrient Load (G_{NL})	15%, $i = 2$
	Turbidity (G_T)	15%, $i = 3$
Fractional Vegetation Coverage (G_{FVC})		10%, $i = 4$
Soil Quality (G_{SQ})	Soil pH (G_{SpH})	15%, $i = 5$
	Organic Matter Content (G_{OMC})	15%, $i = 6$
Altitude (G_A)		15%, $i = 7$

Table 4. Branching Factors of Human Activities

Factors	Weighting (γ_i)
Population Density (H_{PD})	30%, $i = 1$
Noise Levels (H_{NL})	40%, $i = 2$
Invasive Species Dominance (H_{ISD})	30%, $i = 3$

Step 3 + 4

First type of normalizing formula:

$$\text{Normalized value} = \frac{x - \text{Min. Value}}{\text{Optimal Value} - \text{Min. Value}} \times 100$$

It is used when **one** of the **extremes** in the scale of grading is the **optimal value** for the survival of species.

Second type of normalizing formula:

$$\text{Normalized value} = 100 \times \left(1 - \left(\frac{|x - \text{Optimal Value}|}{D(x)}\right)^2\right)$$

It is used when the **optimal value** for the survival of species is **within** the **two extremes** of the scale.

D(x) is the **adaptive normalization range**:

$$D(x) = \text{Min. value} \times (1 - S(x - \text{Optimal value})) + \text{Difference} \times S(x - \text{Optimal Value})$$

$$\text{Difference} = \text{Max. value} - \text{Min. value}$$

S(x) is the **sigmoid function**:

$$S(x) = \frac{1}{1 + e^{-kx}},$$

$$\text{where } k = \frac{5}{\text{Distance between optimal and max. value}}$$

The mathematical expression for **climatic indicators** in the PEB model equation has a form of

$$C_n = \alpha_1 C_{nPA} + \alpha_2 C_{nPV} + \alpha_3 C_{nAQI} + \alpha_4 C_{nH} + \alpha_5 C_{nSI} + \alpha_6 C_{nMT} + \alpha_6 C_{nSI} + \alpha_7 C_{nSF},$$

where C_{nx} is the normalized data of factor C_x.

The mathematical expression for **geographic indicators** in the PEB model equation has a form of

$$G_n = \beta_1 G_{nCDO} + \beta_2 G_{nNL} + \beta_3 G_{nT} + \beta_4 G_{nFVC} + \beta_5 G_{nSpH} + \beta_6 G_{nOMC} + \beta_7 G_{nA},$$

where G_{nx} is the normalized data of factor G_x.

The mathematical expression for **human activities indicators** in the PEB model equation has a form of

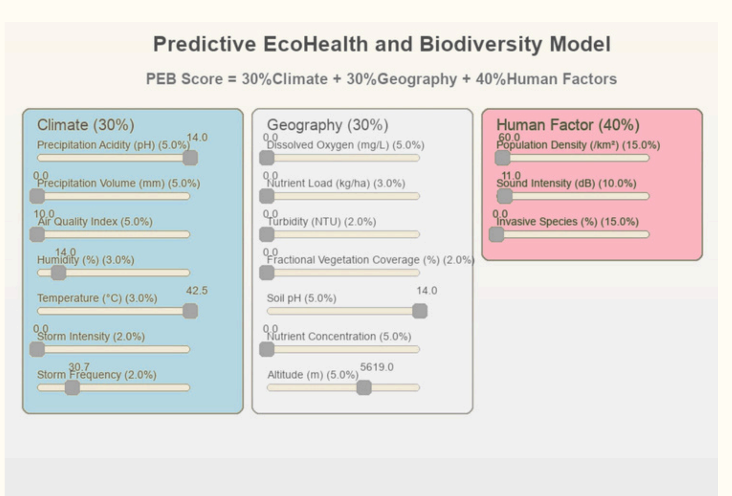
$$H_n = \gamma_1 H_{nPD} + \gamma_2 H_{nNL} + \gamma_3 H_{nISD}$$

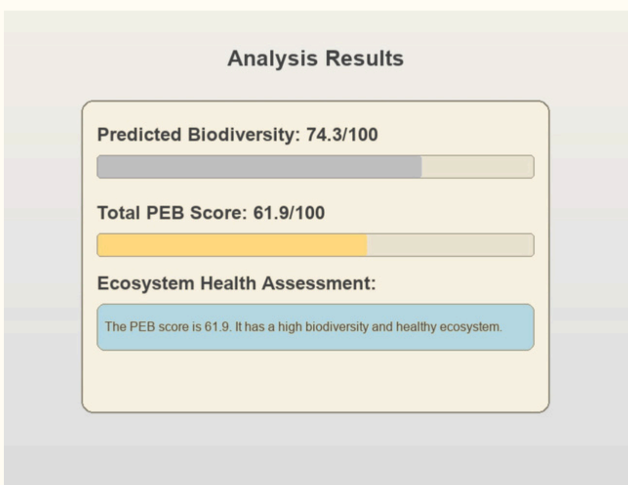
where H_{nx} is the normalized data of factor H_x.

Final Equation in the PEB model

$$PEB_{score} = 30\%\ C_n + 30\%\ G_n + 40\%\ H_n$$

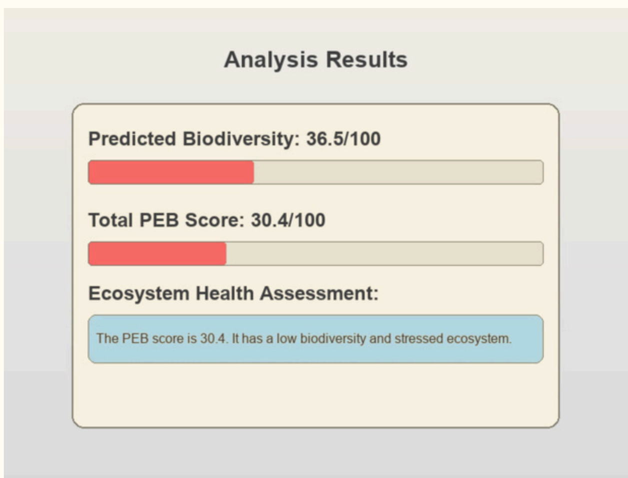
Step 4





Determine the result using Python coding and visualization

Full PBE model

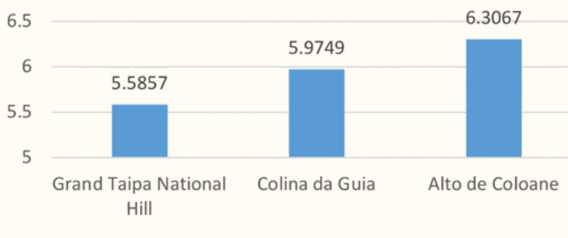


ASSESSING BIODIVERSITY OF MOUNTAINS BY CALCULATING BIODIVERSITY INDEXES

Shannon-Wiener Diversity Index (H'):

a measure of **diversity** which considers the **number of species present**, as well as the **relative abundance** of each species.

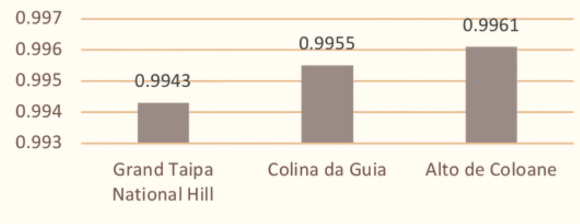
$$H = -\sum_{i=1}^{s} (P i \ln P i)$$



Simpson Diversity Index (D):

a metric used to quantify the **diversity within a community**, which considers both the number of different species (**richness**) and their relative abundance (**evenness**) in a sample.

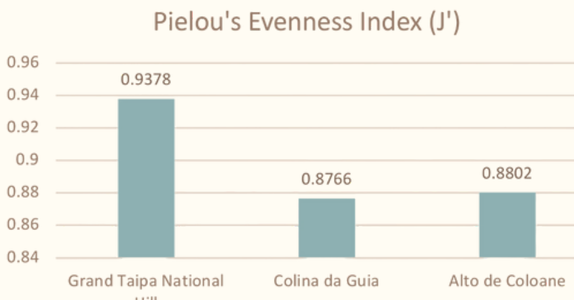
$$D = \left(1 - \frac{\Sigma^n (n-1)}{N(N-1)}\right)$$



Pielou's Evenness Index (J'):

It builds upon the Shannon Diversity Index (H'). a measure of how **evenly species are distributed** in a community.

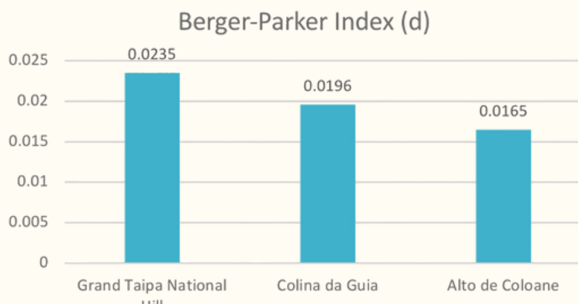
$$J' = \frac{H'}{\ln(S)}$$

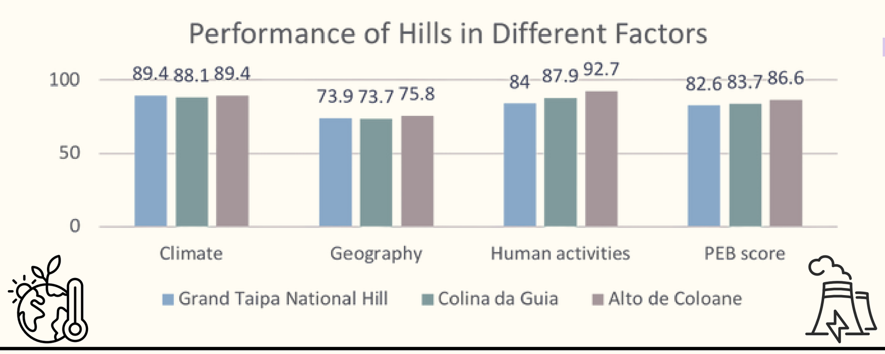


Berger-Parker Index (d):

simple measure of biodiversity that quantifies the **dominance of the most abundant species** in a community.

$$d = \frac{N_{max}}{N}$$





APPLICATION OF MODEL

TRADITIONAL VS. PEB MODEL

Traditional Approach	PEB Model
6–12 months data collection	Real-time assessment (results in 48 hours)
High annual monitoring cost	90% cost reduction by automating analysis
Generalized conservation plans	Precision-targeted interventions (e.g., "Boost Colina's score by addressing Noise Factor 7.3")
Reactive policies (post-degradation)	Proactive urban planning (e.g., rejecting projects in Alto's buffer zone if PEB < 80)

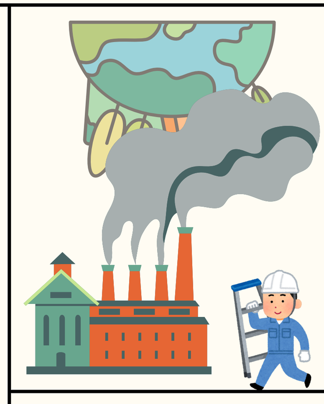
DISCUSSION AND CONCLUSION

Our Predictive EcoHealth and Biodiversity (PEB) model shows:

- **Alto de Coloane** scores **highest** due to:
  - Optimal rainfall, elevation, and waterways
  - Low human disturbance (remote location, adjacent protected areas
- **Colina da Guia** faces **moderate degradation** due to:
  - Urban encroachment (businesses, schools, roads)
- **Grand Taipa National Park** scores low due to:
  - Airport proximity (noise, pollution)

Critical Insight: **Human** activities **degrade ecological health more** powerfully than favorable **climate/geography** can compensate for.

Urgent actions needed:
1) **Urban sprawl** controls
2) **Eco-corridor** protection
3) **Visitor management** in sensitive zones



BIODIVERSITY ANALYZER: PYTHON CODING

To quantify biodiversity patterns, we created a **Python-based Biodiversity Analyzer** that processes iNaturalist observational data.

The tool automates four core functions:
1. spatial **filtering** of study area datapoints
2. **computation** of ecological **indices** (Simpson Diversity, Shannon-Wiener Diversity, Pielou's Evenness, and Berger-Parker Dominance)
3. **taxonomic ranking** of dominant species
4. **specialized filtering** for indicator, rare, and dominant species. This automated pipeline ensures reproducible, error-free analysis while the multi-index approach offers comprehensive ecosystem health assessments across sites.

Full coding

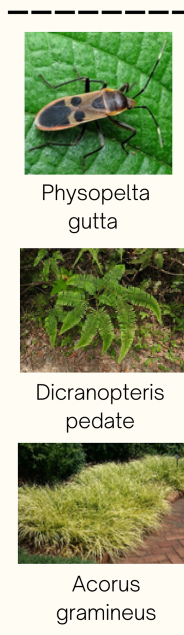
Example result

DOMINANT SPECIES, RARE SPECIES AND ECOLOGICAL INDICATOR SPECIES

Colina da Guia (Guia Hill)

Features: 90m-high, **Macau Peninsula's highest point** with the iconic Guia Lighthouse (UNESCO site) and Chapel of Our Lady of Guia

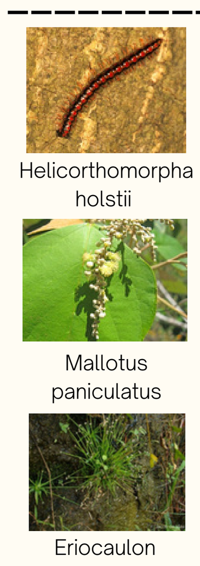
Dominant species	Rare species	Indicator species
Spilopelia chinensis (Count: 90, 1.96%), **Passer montanus** (Count: 68, 1.48%)	**Casuarina equisetifolia** (Count: 19, 0.415%), **Pterocarpus indicus** (Count: 2, 0.0436%)	**Lissachatina fulica** (Count: 17, 0.371%), **Paratrechina longicornis** (Count: 22, 0.480%)



Alto de Coloane (Coloane Hill)

Features: 171m-high, it's **Macau's highest natural peak**, home to the A-Ma Cultural Village and the world's tallest white jade Mazu statue

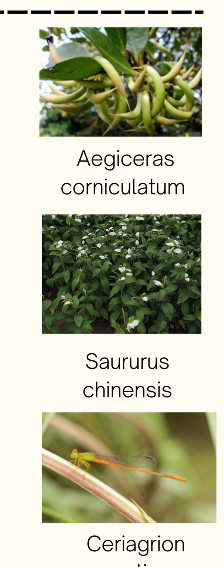
Dominant species	Rare species	Indicator species
Physopelta gutta (Count: 84, 1.55%), **Dicranopteris pedate** (Count: 89, 1.65%)	**Acorus gramineus** (Count: 5, 0.0926%), **Araucaria heterophylla** (Count: 2, 0.0370%)	**Papilio polytes** (Count: 8, 0.148%), **Mussaenda formosanum** (Count: 6, 0.111%)



Grand Taipa National Hill (Taipa Hill)

Features: 159m-high, the **tallest hill in Taipa**, with hiking trails overlooking Cotai and the airport wetlands

Dominant species	Rare species	Indicator species
Helicorthomorpha holstii (Count: 19, 2.35%), **Mallotus paniculatus** (Count: 15, 1.85%)	**Eriocaulon truncatum** (Count: 1, 0.124%), **Aegiceras corniculatum** (Count: 1, 0.124%)	**Saururus chinensis** (Count: 1, 0.124%), **Ceriagrion auranticum** (Count: 4, 0.494%)



ENVIRONMENTAL PROTECTION-SUGGESTIONS

Our findings highlight **urgent steps** to safeguard Macau's mountain **biodiversity** and **ecohealth**.

Words to citizens:

As our data shows these stressors degrade habitats,
- **Respect Buffer Zones**: Avoid venturing beyond marked trails to prevent habitat fragmentation
- **Contribute to iNaturalist**: Photograph species during visits to strengthen monitoring data
- **Carry-In/Carry-Out**: Remove all waste – even biodegradable items disrupt soil chemistry
- **Visit Off-Peak**: Choose weekday mornings to reduce cumulative foot traffic
- **Practice Quiet Recreation**: Use headphones instead of speakers to minimize noise pollution
- **Join "Trail Guardians"**: Volunteer for monthly habitat restoration (e.g., invasive species removal)

Suggestions to Macao government:

Policy Recommendation: Adopt the PEB Model for Efficient Biodiversity Governance
- **traditional methods** demand costly and multi-year data collection
- → **delaying critical conservation interventions**
- **our model is a perfect solution**
- strongly recommend Macau's government **implement our Predictive EcoHealth and Biodiversity (PEB) model**
- **- make proactive actions**
- **- conservation planning**

such as rejecting site development projects in ecologically sensitive zones (e.g., Alto de Coloane buffer areas with PEB scores <80)

LIMITATIONS

1. **Data Accuracy from the App**
   - **Automated** species identification may **misclassify** species.
   - **Similar-looking** species might be **indistinguishable**, **underestimating** biodiversity.
2. **Sampling Bias and Lacking Data**
   - Contestants may favor certain **accessible areas**, **skewing** data representation.
   - **Less visible** species (e.g., nocturnal animals, small insects) may be overlooked.
   - **Certain groups** (e.g., fungi, microorganisms) may be **underrepresented**.
   - **Inaccurate metadata** (e.g., GPS, timestamps) reduces **reliability** of analyses.
3. **Duplicate Records**
   - **Multiple photos** of the same organism may **inflate species counts**, leading to **overestimation** of biodiversity.
4. **Environmental Variability**
   - **Weather, time of day**, or **seasonality** can affect species **visibility** and **consistency** of data.
   - **Human disturbance** may alter species behavior or **distribution** temporarily.
5. **Resource Constraints**
   - **Limited time** and **manpower** restrict ecological modeling and validation.
   - **Advanced analyses** may be hindered by **computational tool constraints**.
6. **Estimation Errors in Analysis**
   - **Statistical models** depend on **assumptions** that **may not be valid**.
   - **Small sample sizes** or **patchy data** can amplify extrapolation **errors**.

REFERENCES